**Paul: James' friend Ben asked me to look at this on his GitHub**

This is better than a toy project. It shows decent structure, some statistical awareness, and unusually good self-critique for a GitHub quant repo. My take from the excerpt is:

**What's good**

- He is combining two genuinely different detectors rather than stacking variants of the same idea. Geometric path shape plus hidden-state Markov is a sensible ensemble idea.
- He is explicitly focused on **risk reduction** rather than pretending to beat buy-and-hold on raw CAGR at all costs.
- He is surfacing weaknesses in the report itself: transaction costs, in-sample thresholds, SPY-only calibration. That is a good sign.
- The code is fairly readable and modular: `data`, `geometric`, `markov`, `ensemble`, `backtest`.

**What jumps out as the main weakness**
The biggest issue is not coding quality. It is **research design leakage / optimistic framing risk**.

From the excerpt:

- thresholds are described as adaptive percentiles,
- ensemble cutoffs appear derived on the full dataset,
- the Markov model is fit on the return history,
- then backtest stats are presented over the same long sample.

Even though the captions say things like "no look-ahead" and "training data only," the actual excerpt does not show a true walk-forward framework. If thresholds, state labeling, or model selection were materially informed by the full sample, then the reported Sharpe is at least partly in-sample.

That is the first place I would push him.

# On the Markov model specifically

This is where I'd be most critical.

A vanilla 3-state Markov model on daily SPY returns often ends up learning something close to:

- up / quiet
- chop
- crisis / negative volatility cluster

That can be useful, but it is also fragile because:

1. **State semantics drift over time.** The “momentum” state in 2003 may not look like the one in 2021.
2. **State labels are not naturally stable.** Hidden-state models can swap interpretation unless you enforce a consistent mapping rule.
3. **Gaussian-ish assumptions can be too weak for equity returns**, especially in crisis tails.
4. **AR(1) on daily returns is usually only mildly informative** for SPY unless the real edge is mostly coming from volatility clustering/crisis filtering rather than directional persistence.
5. **Filtered probabilities can still look clean in hindsight** even when real-time classification is noisier.

# My suggestions to improve the Markov piece

## 1) Make the Markov model explain a richer observation vector

Right now it looks like it is fit on returns, maybe AR(1). I would strongly prefer the hidden state to infer from a multivariate observation set such as:

- daily return
- rolling realized vol
- downside semivol
- gap size
- range expansion / contraction
- distance from moving average or trend slope
- maybe volume/turnover regime

That usually makes the states more economically interpretable.

## 2) Use a regime model aimed more at volatility/state persistence than pure directional prediction

For equities, the Markov edge is often strongest as a **risk-state filter**, not as a direct long/flat timing engine. So I’d ask:

- is the Markov model really adding directional information?
- or is it mostly identifying “don’t be aggressive here” periods?

That distinction matters.

## 3) Enforce state labeling rules explicitly

Do not let the model decide after the fact which state is “momentum” or “crisis” based on eyeballing charts.
Map states mechanically each refit, for example:

- highest mean / moderate vol = momentum
- near-zero mean / low trendiness = choppy
- lowest mean / highest vol = crisis

Without that, comparisons across time are shaky.

### 4) Use walk-forward refits

This is the big one.

He should do something like:

- initial training window
- fit model
- classify next block out of sample
- roll forward
- refit
- repeat

And do the same for:

- percentile thresholds
- ensemble cutoffs
- any crisis override threshold

That would make the Sharpe much more believable.

### 5) Compare against simpler baselines

He needs to show the Markov adds value beyond very simple filters:

- 200d MA long/cash
- realized vol filter
- drawdown filter
- trend + vol gate
- plain geometric detector alone

If the Markov model does not beat those on a clean walk-forward basis, it is probably adding complexity more than edge.

### 6) Test probability calibration, not just regime prettiness

A lot of regime plots look convincing. The question is whether:

- high P(crisis) really predicts weaker forward returns or higher downside tail risk,
- high P(momentum) really predicts better forward risk-adjusted returns,
- and whether those relationships persist out of sample.

Reliability curves / decile tests would help.

### 7) Check robustness to state count

If k=3 was chosen by BIC, fine, but he should still show:

- k=2
- k=3
- k=4

If the conclusions collapse when moving from 3 to 4 states, then the interpretation may be overfit.

## Specific things I noticed in the code/report excerpt

A few direct observations:

### Report/page assembly bug or at least awkwardness

He saves pages in this order:

- make_page3
- make_page1
- make_page2

and the footers say:

- page 1 of 3 on make_page3
- page 2 of 3 on make_page1
- page 3 of 3 on make_page2

So it is internally consistent, but confusing in code and maintenance. I would clean that up.

### "Breakeven cost" calculation looks suspicious

He says:

```
breakeven_bps = int(_bps_range[_np.argmin(_np.abs(_np.array(_cost_sharpes)))])
```

That is not really “Sharpe = 0 crossover.” It is just the sampled bps whose Sharpe is **closest** to zero over 0–30 bps. That is a rough approximation, not an actual crossover estimate.

Better:

- interpolate between the two neighboring points around zero,
- or solve on a finer grid.

### Transaction cost treatment may still be too light

For a switching strategy on daily SPY:

- round-trip bps alone may understate reality,
- especially if execution lag, slippage, and opening/closing mechanics matter.

He does mention 1-day execution lag, which is good, but I would still want:

- open-to-close vs close-to-close consistency,
- explicit turnover numbers,
- exact cost application logic per position change.

### Mixed regime being the “most significant signal”

Interesting finding, but I would be careful. “Mixed” may just be a bucket where:

- equity risk premium keeps bleeding upward,
- while the detectors disagree often enough that it looks statistically strong.

That can be real, but it can also be an artifact of how the ensemble is partitioning states. Needs out-of-sample confirmation.

### T-stats on daily returns

Useful as a quick diagnostic, but not enough. Daily returns are heteroskedastic and autocorrelation assumptions can get messy. I would want:

- Newey-West adjusted inference,
- block bootstrap,
- or at least bootstrap confidence intervals.

## Bottom line assessment

My honest verdict:

**Promising and thoughtful, but not yet institutionally convincing.**

It looks like a smart exploratory research project, not nonsense.
But the key hurdle is whether the edge survives:

- true walk-forward fitting,
- cleaner cost modeling,
- baseline comparisons,
- and stability testing of the Markov state definitions.

If it survives that, then it becomes interesting. If not, the Markov component may be mostly giving an elegant narrative around a crisis-avoidance filter.

## What I would tell Ben in one paragraph

You could say something like this:

Message

Took a look — overall I think it's a solid piece of exploratory research and better than most GitHub quant repos, especially in how clearly you lay out the limitations. My main pushback is less on coding and more on research design: I'd want to see a stricter walk-forward framework for the Markov fit, thresholds and ensemble cutoffs, plus comparisons against much simpler baselines like trend/vol filters to prove the Markov piece is adding real value rather than mostly acting as a dressed-up crisis filter. On the Markov side specifically, I'd consider richer observations than just returns/AR(1), enforce consistent state labeling at each refit, and test robustness across k=2/3/4 and with probability calibration rather than just nice-looking regime plots. If it still holds up after that, then it starts to look genuinely compelling.

If you want, I can also turn this into a more technical review for him with numbered recommendations and a few code-level suggestions.

**Paul: insightful we could suggest rather than AR(1)? And maybe something (paradoxically) more basic such as long term trend vs short term possibly as simple as 200day vs 30day MA?**

Yes — both of those are good instincts.

For this kind of daily equity regime model, **AR(1) is often too weak and too literal**. It asks "are returns serially correlated one step ahead?" when the more useful question is usually:

- is the market in a **trend / anti-trend / crisis-volatility** environment?
- is persistence happening in **price**, **volatility**, or **drawdown behavior** rather than in raw daily returns?

So I would suggest two upgrades, one “smarter” and one “simpler.”

# 1) More insightful than AR(1)

Instead of a Markov model on raw daily returns with AR(1), I’d look at a hidden-state model driven by **features that actually define regime**.

### Better observation set for the Markov model

Use a multivariate observation vector such as:

- 20d or 30d return
- 5d return
- realized volatility
- downside semivol
- drawdown from 252d high
- distance from 200d MA
- slope of 50d or 100d MA
- maybe return autocorrelation or straightness ratio

That lets the hidden states represent things like:

- **trend-up / orderly**
- **trend-up but stressed**
- **sideways/chop**
- **downtrend / crisis**

That is much more useful than asking AR(1) to infer all of that from one lagged return.

### Even better framing

For equities, I’d often rather have the hidden model explain:

- **trend strength**
- **volatility state**
- **stress/crash probability**

than raw direction.

So a stronger version might be:

- Hidden Markov Model or Markov-switching model on **return + vol + trend features**
- or even a **semi-Markov** style persistence rule if he wants states not to flicker too much

The real edge is often in detecting:

- “long-only is safe”
- “long-only but smaller”
- “get defensive”

not in predicting tomorrow’s sign of return.

## 2) Paradoxically better because simpler

Yes — something like **long-term trend vs short-term trend** can be very powerful, and often more robust.

A very clean regime map could be:

- **200d MA slope or price vs 200d MA** = long-term backdrop
- **30d MA slope or price vs 30d MA** = short-term posture

That gives a simple 2x2 framework:

### Regime grid

- **Above 200d, above 30d** → strong trend / momentum
- **Above 200d, below 30d** → pullback within bull trend
- **Below 200d, above 30d** → rally in weak backdrop / transition
- **Below 200d, below 30d** → weak / defensive regime

That is intuitive, stable, and actually economically meaningful.

## Why this may be better than AR(1)

Because equities tend to have:

- weak short-horizon return autocorrelation,
- stronger **medium-term trend persistence**,
- and very strong **volatility clustering / crash clustering**.

So:

- AR(1) on daily returns often captures very little,
- while 30d vs 200d captures the thing you actually care about: **state of market structure**.

## A very practical suggestion for Ben

I'd suggest he test three versions side by side:

### A. Very simple baseline

Use only:

- price relative to 200d MA
- price relative to 30d MA

Maybe score it like:

- +1 if above 200d
- +1 if 30d slope positive
- +1 if price above 30d
  Then map high score = momentum, low score = defensive.

### B. Slightly richer but still simple

Use:

- 200d trend
- 30d trend
- realized vol
- drawdown from 252d high

This may outperform the Markov AR(1) while being easier to explain.

### C. Improved hidden-state version

If he wants to keep the Markov architecture, feed it:

- 20d return
- 5d return
- 20d realized vol
- drawdown
- distance from 200d MA

That would be a much more meaningful hidden-state model.

## The most interesting hybrid

My guess is the best design may be:

- **Simple structural regime backbone**:

    - long-term trend = 200d
    - short-term trend = 30d
- **Markov overlay only for stress/crisis filtering**:
    - hidden state used mainly to detect elevated crash/stress probability

That way:

- the simple MA structure defines the main regime,
- the Markov component is not asked to do too much,
- and the overall model becomes easier to trust.

That is often better than making the Markov model the star of the show.

## If you want one crisp recommendation to send him

You could say:

Message

One thought: AR(1) on raw daily returns may be too narrow for what you actually want the hidden-state model to detect. For equities I'd consider either a simpler structural regime model using long-term vs short-term trend (e.g. 200d vs 30d MA, or price/slope relative to those) or, if you keep the Markov piece, feed it a richer observation vector like 5d/20d returns, realized vol, drawdown, and distance from the 200d rather than just lagged returns. My intuition is that a 200d/30d trend backbone plus a Markov-style stress filter might end up more robust and interpretable than a pure AR(1) hidden-state setup.

A slightly sharper version, if you want, is that **AR(1) is trying to model tomorrow's return, whereas regime models should really model market state**.

**Paul: Yes where market state is actually measurable at least as one defines it and that state correlates to direction of movement especially when informed by a 200/30 MA trend grid and could be augmented with VIX, RVol and possibly volume at price and daily volume**

Exactly. That is the stronger framing.

The key shift is:

**don't ask the model to discover "regime" from weak daily return autocorrelation alone; define market state using observable state variables that are plausibly causal or at least structurally related to future drift and risk.**

That is a much more defensible research program.

# Better conceptual model

Instead of:

- hidden state inferred mainly from AR(1) on returns

use:

- **state = function of trend, stress, participation, and location**

where for SPY or index equities those can be approximated by:

- **trend**: 200d and 30d MA grid
- **stress**: VIX, realized vol, downside vol, drawdown
- **participation**: daily volume, breadth if available
- **location/context**: distance from 200d, distance from recent range extremes, maybe volume-at-price / acceptance zone measures

Then ask:

- when those state variables line up a certain way, what is the conditional forward return distribution?
- and does that produce a stable long / half-long / cash mapping?

That is much more insightful.

# A strong practical regime framework

You could almost define a regime engine in layers.

### Layer 1: Structural trend regime

Very basic and robust:

- Price vs 200d MA
- 200d MA slope
- Price vs 30d MA
- 30d MA slope

This gives the backbone.

For example:

- **Bull impulse**: above 200d, 200d rising, above 30d, 30d rising
- **Bull pullback**: above 200d, 200d rising, below 30d or 30d flattening
- **Bear rally / transition**: below 200d but above 30d
- **Bear stress**: below 200d, 200d falling, below 30d

Already this is more interpretable than AR(1).

## Layer 2: Stress / fragility overlay

This is where VIX and realized vol belong.

Useful variables:

- VIX level
- VIX term structure if available
- 20d realized vol
- downside semivol
- drawdown from 252d high
- vol-of-vol or recent change in vol

This lets you distinguish:

- healthy bull
- stressed bull
- orderly correction
- unstable/crisis regime

That is extremely important because “above the 200d” is not enough if vol is exploding.

## Layer 3: Participation / confirmation

This is where volume helps.

Useful variables:

- daily volume vs 20d average volume
- up-volume vs down-volume
- breadth, if he can get it
- maybe OBV or accumulation/distribution style measures

These help answer:

- is the move broadly participated in?
- is the trend thinning out?

- is the reversal real or just a squeeze?

### Layer 4: Location / acceptance

This is the more nuanced piece, and where your instinct about volume at price is interesting.

Possible variables:

- distance from 200d MA in z-score terms
- distance from 20d range high/low
- drawdown from recent high
- volume-at-price concentration / HVN-LVN type structure if available
- whether price is breaking out of balance or reverting into value

For daily SPY, full VAP may be a bit elaborate unless he has a clean implementation, but the idea is sound:

- markets behave differently when they are trending away from value than when they are rotating within value.

## Why this is better than AR(1)

Because daily equity regimes are not really about one-lag autocorrelation.

They are more about:

- **trend persistence**
- **volatility clustering**
- **stress cascades**
- **participation**
- **location relative to accepted value and long-term trend**

AR(1) can only weakly proxy that. Your proposed framework measures it more directly.

## Best way to formalize it

I would suggest Ben think in terms of a **state vector**, not a single model family.

For example:

$$S_t = \{\text{LT trend, ST trend, vol regime, participation, location}\}$$

Then either:

### Option A: Explicit rule-based regime map

Very robust and interpretable.

Example:

- if above 200d and 200d slope positive and VIX moderate and RVOL not stressed → momentum regime
- if above 200d but below 30d and VIX rising → pullback / mixed regime
- if below 200d and VIX high and drawdown expanding → defensive / crisis regime

This is often underrated.

### Option B: Hidden-state model on the state vector

More sophisticated.

Feed the hidden-state model variables like:

- 30d return
- distance from 200d
- 20d realized vol
- VIX
- volume surprise
- drawdown

That is a much better hidden-state input than AR(1) daily return.

## My honest view on volume-at-price

It is conceptually attractive, but I would be cautious for this project.

For a daily SPY regime model:

- 200/30 structure, VIX, RVOL, drawdown, and daily volume are probably the first-order gains.
- Volume-at-price may add value, but it can also add complexity and implementation ambiguity faster than signal.

So I would rank the improvements:

1. 200d / 30d trend grid

2. VIX + realized vol + drawdown
3. Daily volume / RVOL
4. Only then experiment with volume-at-price / value-area style context

Unless he already has a very clean VAP framework.

## What the state should predict

Crucially, he should not force state to predict only direction.

State should predict at least three things:

- expected forward return
- expected volatility / drawdown risk
- confidence in being long

So the output should maybe be:

- **full long**
- **half long**
- **cash**
  rather than long/cash based purely on directional sign.

That matches his stated aim of risk reduction.

## A clean research design for him

I'd suggest this test ladder:

### Model 1 — Purely simple

Regime defined by:

- price vs 200d
- 30d slope
- VIX bucket

### Model 2 — Simple plus stress

Add:

- realized vol
- drawdown from 252d high

- daily volume vs 20d average

### Model 3 — Hidden-state on state variables

HMM or Markov-switching model on:

- 20d return
- distance from 200d
- 20d realized vol
- VIX
- RVOL

Then compare all three out of sample.

My suspicion is:

- Model 1 will be surprisingly strong,
- Model 2 may be the sweet spot,
- Model 3 may add only modest improvement unless very carefully done.

## Best concise phrasing to send him

Message

Yes — I think the stronger approach is to define market state using variables that are actually measurable and economically meaningful, rather than asking AR(1) on daily returns to infer regime indirectly. For equities, I'd think in terms of a state vector built from a 200d/30d trend grid, stress variables like VIX / realized vol / drawdown, participation measures like daily volume or RVOL, and possibly location/context variables such as distance from the 200d or value-area style measures. Then either map those explicitly into momentum / mixed / defensive regimes, or feed that richer state vector into a hidden-state model. My instinct is that a simple 200/30 trend backbone plus a stress overlay may end up both more robust and more interpretable than a pure AR(1)-style Markov setup.

The really sharp line is this:

**state should be defined by market structure, not inferred mainly from return autocorrelation.**

**Paul: What about VVIX/VIX and GEX? He can pull GEX data from Squeezemetrics**
Thought for 15s

Yes — **VVIX/VIX and GEX are probably more relevant than AR(1)** if the goal is to describe equity **market state** rather than one-day return autocorrelation.

Squeezemetrics explicitly defines **GEX** as a measure of dealers' hedging obligations from existing SPX options: positive GEX implies hedging that tends to **dampen volatility** by buying into weakness and selling into strength, while negative GEX implies the opposite, tending to **magnify volatility**. Their documentation also frames GEX as one of the core axes for reading market state.

On the volatility side, **VIX** is Cboe's measure of the market's expectation of **30-day forward volatility** for the S&P 500, inferred from SPX option prices, and **VVIX** is the analogous volatility measure derived from **VIX option** prices—effectively the market's expectation of volatility in volatility.

So conceptually, a strong hierarchy would be:

- **Trend backbone:** 200d / 30d MA grid
- **Stress overlay:** VIX, VVIX, realized vol, drawdown
- **Dealer-flow overlay:** GEX
- **Participation overlay:** daily volume / RVOL
- **Optional refinement:** value/acceptance proxies, maybe volume-at-price if implemented carefully

That is a much more market-structural definition of regime.

The useful distinction is:

- **VIX** tells you how much index volatility is being priced.
- **VVIX** tells you how unstable the volatility regime itself may be.
- **GEX** tells you whether dealer hedging is more likely to act as a **shock absorber** or a **shock amplifier**.

That combination is powerful because you can have very different states that AR(1) would miss, for example:

- bullish trend + positive GEX + moderate VIX = orderly grind
- bullish trend + negative GEX + high VVIX = unstable squeeze / air pockets possible
- bearish backdrop + negative GEX + rising VIX = fragile, potentially reflexive downside
- above 200d but VVIX elevated = trend may still be up, but path risk is worsening

One nuance: Squeezemetrics also noted that when GEX goes negative, dealer hedging becomes more of a **multiplier on volume** than an independent trigger, so negative GEX should not be treated as a standalone directional signal; it is more a fragility/convexity condition.

So I would not use GEX as "buy when high, sell when low." I would use it more like:

- **positive GEX** → mean reversion / lower realized vol more plausible
- **negative GEX** → breakout, acceleration, and gap-extension risk higher

- **VVIX rising fast** → regime instability increasing, even if spot VIX is not yet extreme

The cleanest research design would be to treat these as **state descriptors**, not direct trading signals. For example:

**State vector**

- LT trend: price vs 200d, 200d slope
- ST trend: price vs 30d, 30d slope
- Vol state: VIX level, VIX trend, VVIX level/change, realized vol
- Flow state: GEX sign and magnitude
- Participation: RVOL / daily volume surprise

Then test conditional forward outcomes:

- next 5d / 20d return
- realized drawdown
- vol-adjusted return
- probability of large move

My guess is that for Ben's project, **200/30 trend + VIX/VVIX + GEX** is a far stronger regime foundation than AR(1), and probably more interpretable too. The likely role of each is:

- **200/30 MA grid:** defines structural trend
- **VIX/VVIX:** defines stress and instability
- **GEX:** defines whether the options market is likely to dampen or amplify moves
- **RVOL:** confirms whether the move has participation

That gives him a regime model that actually matches how the market behaves.

A concise message you could send him:

Message

Yes — I think VVIX/VIX and GEX are much more relevant regime variables than AR(1) on daily returns. A 200d/30d MA grid can define the structural trend state, VIX and especially VVIX can describe stress / instability, and GEX can describe whether dealer hedging is more likely to dampen or amplify moves. That feels closer to an actual market-state model: trend, stress, convexity, participation. I'd be inclined to use those as explicit state variables first, and only then decide whether a hidden-state model still adds value.

One caution: I would keep **GEX** as an overlay on **SPX/SPY index-state work**, but be more careful applying it as a generic signal to single names or assuming precise directional timing from it.

**Paul: Can you give me a pdf of our chat with my questions and your responses that I can send to Ben so he sees the evolution of thoughts and suggestions are then in context?**

**PDF was weak so better to share the whole thread above.**